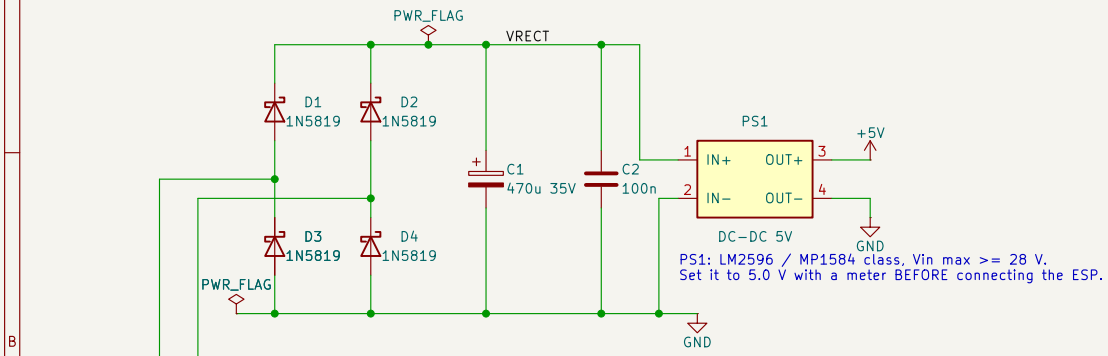
1. Track input and power

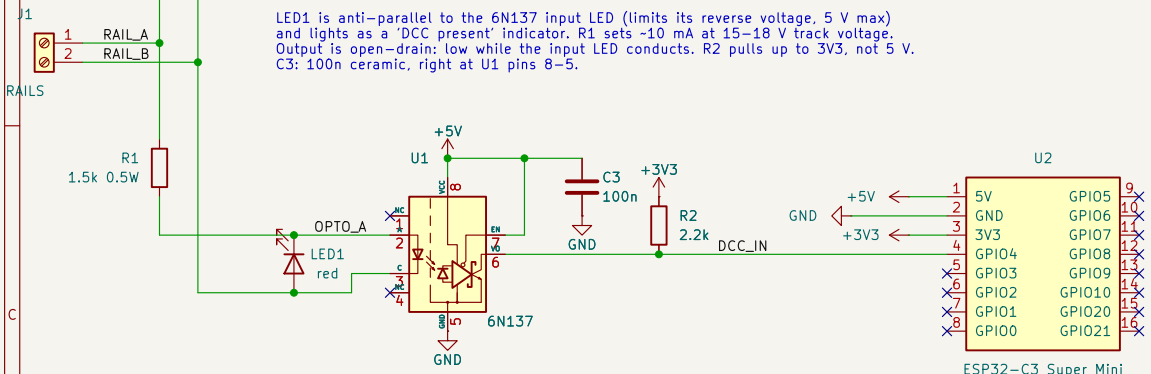
DCC from the rails is a bipolar square wave: rectify, buffer, step down to 5 V.
C1 also carries the board through each RailCom cutout (~0.5 ms, rails at 0 V).
IN− and OUT− of a non-isolated module are the same node, so GND is tied to the rails via the bridge: never plug in USB while powered from the rails.



PS1: LM2596 / MP1584 class, Vin max >= 28 V.
Set it to 5.0 V with a meter BEFORE connecting the ESP.

2. DCC signal input (6N137 optocoupler)

LED1 is anti-parallel to the 6N137 input LED (limits its reverse voltage, 5 V max)
and lights as a 'DCC present' indicator. R1 sets ~10 mA at 15–18 V track voltage.
Output is open-drain: low while the input LED conducts. R2 pulls up to 3V3, not 5 V.
C3: 100n ceramic, right at U1 pins 8–5.



3. ESP32-C3 Super Mini

GPIO4 = DCC_IN (firmware: main/node_config.h)
GPIO8 onboard LED, GPIO9 BOOT,
GPIO2/8/9 strapping pins: keep free.

Rail-powered ESP32-C3 Super Mini, 6N137 DCC input on GPIO4
labs/dcc
Sheet: /
File: dcc-sniffer.kicad_sch

Title: DCC sniffer / accessory decoder input stage

Size: A4	Date: 2026-11-01	**Rev: 2**
KiCad E.D.A. 10.0.6		Id: 1/1